Choice-free descent of functors along weak equivalences into univalent categories

The Univalent Foundations Program

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Editorial note. Complete author proof and local natural-transformation descent are included. This migrates S046-ct:cat-weq-eq once; supporting functor-category univalence is not an extra counted task.

Editorial note. The word category means the book univalent-category condition, not an arbitrary set-based category. Essential surjectivity is truncated and no global choices of object lifts are assumed.

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Editorial note. 2026-10-11 材料收口：复用既有 H2、来源许可和近邻判断；核对原命题与作者证明起止、所指上下文及装配范围齐备。未重证作者结论；沿用既有非重复依据。

Allowed standard prerequisites (editorial).

- Basic univalent type theory and elementary category equivalence. Function extensionality, dependent-pair path transport and contractibility as a mere proposition; propositional truncation may be eliminated only into mere propositions. The elementary fully-faithful plus split-essential-surjectivity criterion for equivalence is allowed; an unsplit essentially-surjective functor is not assumed to have a chosen inverse. Functor-category univalence and local precomposition fullness are supplied with original proofs.

Source reference. Original lemma 9.4.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1100–1100. Explicit standard equivalence criterion with specified object lifts; it does not supply the target unsplit descent construction, whose contractible X_b/Y_f proofs are retained.

Definition 9.9.1. A **precategory** A consists of the following.

(i) A type A_0, whose elements are called **objects**. We write $a : A$ for $a : A_0$.

(ii) For each $a, b : A$, a set $\hom_A(a, b)$, whose elements are called **arrows** or **morphisms**.

(iii) For each $a : A$, a morphism $1_a : \hom_A(a, a)$, called the **identity morphism**.

(iv) For each $a, b, c : A$, a function

$$\hom_A(b, c) \to \hom_A(a, b) \to \hom_A(a, c)$$

called **composition**, and denoted infix by $g \mapsto f \mapsto g \circ f$, or sometimes simply by gf.

(v) For each $a, b : A$ and $f : \hom_A(a, b)$, we have $f = 1_b \circ f$ and $f = f \circ 1_a$.

(vi) For each $a, b, c, d : A$ and

$$f : \hom_A(a, b), \qquad g : \hom_A(b, c), \qquad h : \hom_A(c, d),$$

we have $h \circ (g \circ f) = (h \circ g) \circ f$.

Definition 9.9.2. A morphism $f : \hom_A(a, b)$ is an **isomorphism** if there is a morphism $g : \hom_A(b, a)$ such that $g \circ f = 1_a$ and $f \circ g = 1_b$. We write $a \cong b$ for the type of such isomorphisms.

Lemma 9.9.3 ($\mathsf{idtoiso}$)**.** *If A is a precategory and $a, b : A$, then*

$$(a = b) \to (a \cong b).$$

Definition 9.9.4. A **category** is a precategory such that for all $a, b : A$, the function $\mathsf{idtoiso}_{a,b}$ from Lemma 9.9.3 is an equivalence.

In particular, in a category, if $a \cong b$, then $a = b$.

Lemma 9.9.5. *For $p : a = a'$ and $q : b = b'$ and $f : \hom_A(a, b)$, we have*

$$(p, q)_*(f) = \mathsf{idtoiso}(q) \circ f \circ \mathsf{idtoiso}(p)^{-1}. \tag{9.9.6}$$

证明. By induction, we may assume p and q are refl_a and refl_b respectively. Then the left-hand side of (9.9.6) is simply f. But by definition, $\mathsf{idtoiso}(\mathsf{refl}_a)$ is 1_a, and $\mathsf{idtoiso}(\mathsf{refl}_b)$ is 1_b, so the right-hand side of (9.9.6) is $1_b \circ f \circ 1_a$, which is equal to f. □

Definition 9.9.7. Let A and B be precategories. A **functor** $F : A \to B$ consists of

(i) A function $F_0 : A_0 \to B_0$, generally also denoted F.

(ii) For each $a, b : A$, a function $F_{a,b} : \hom_A(a, b) \to \hom_B(Fa, Fb)$, generally also denoted F.

(iii) For each $a : A$, we have $F(1_a) = 1_{Fa}$.

(iv) For each $a, b, c : A$ and $f : \hom_A(a, b)$ and $g : \hom_A(b, c)$, we have

$$F(g \circ f) = Fg \circ Ff.$$

Note that by induction on identity, a functor also preserves idtoiso.

Definition 9.9.8. For functors $F, G : A \to B$, a **natural transformation** $\gamma : F \to G$ consists of

(i) For each $a : A$, a morphism $\gamma_a : \hom_B(Fa, Ga)$ (the "components").

(ii) For each $a, b : A$ and $f : \hom_A(a, b)$, we have $Gf \circ \gamma_a = \gamma_b \circ Ff$ (the "naturality axiom").

Since each type $\hom_B(Fa, Gb)$ is a set, its identity type is a mere proposition. Thus, the naturality axiom is a mere proposition, so identity of natural transformations is determined by identity of their components. In particular, for any F and G, the type of natural transformations from F to G is again a set.

Similarly, identity of functors is determined by identity of the functions $A_0 \to B_0$ and (transported along this) of the corresponding functions on hom-sets.

Definition 9.9.9. For precategories A, B, there is a precategory B^A, called the **functor precategory**, defined by

- $(B^A)_0$ is the type of functors from A to B.
- $\hom_{B^A}(F, G)$ is the type of natural transformations from F to G.

证明. We define $(1_F)_a :\equiv 1_{Fa}$. Naturality follows by the unit axioms of a precategory. For $\gamma : F \to G$ and $\delta : G \to H$, we define $(\delta \circ \gamma)_a :\equiv \delta_a \circ \gamma_a$. Naturality follows by associativity. Similarly, the unit and associativity laws for B^A follow from those for B. □

Lemma 9.9.10. *A natural transformation $\gamma : F \to G$ is an isomorphism in B^A if and only if each γ_a is an isomorphism in B.*

证明. If γ is an isomorphism, then we have $\delta : G \to F$ that is its inverse. By definition of composition in B^A, $(\delta\gamma)_a \equiv \delta_a\gamma_a$ and similarly $(\gamma\delta)_a \equiv \gamma_a\delta_a$. Thus, $\delta\gamma = 1_F$ and $\gamma\delta = 1_G$ imply $\delta_a\gamma_a = 1_{Fa}$ and $\gamma_a\delta_a = 1_{Ga}$, so γ_a is an isomorphism.

Conversely, suppose each γ_a is an isomorphism, with inverse called δ_a, say. We define a natural transformation $\delta : G \to F$ with components δ_a; for the naturality axiom we have

$$Ff \circ \delta_a = \delta_b \circ \gamma_b \circ Ff \circ \delta_a = \delta_b \circ Gf \circ \gamma_a \circ \delta_a = \delta_b \circ Gf.$$

Now since composition and identity of natural transformations is determined on their components, we have $\gamma\delta = 1_G$ and $\delta\gamma = 1_F$. □

The following result is fundamental.

Theorem 9.9.11. *If A is a precategory and B is a category, then B^A is a category.*

证明. Let $F, G : A \to B$; we must show that $\mathsf{idtoiso} : (F = G) \to (F \cong G)$ is an equivalence.

To give an inverse to it, suppose $\gamma : F \cong G$ is a natural isomorphism. Then for any $a : A$, we have an isomorphism $\gamma_a : Fa \cong Ga$, hence an identity $\mathsf{isotoid}(\gamma_a) : Fa = Ga$. By function extensionality, we have an identity $\bar{\gamma} : F_0 =_{(A_0 \to B_0)} G_0$.

Now since the last two axioms of a functor are mere propositions, to show that $F = G$ it will suffice to show that for any $a, b : A$, the functions

$$F_{a,b} : \hom_A(a, b) \to \hom_B(Fa, Fb) \qquad \text{and}$$
$$G_{a,b} : \hom_A(a, b) \to \hom_B(Ga, Gb)$$

become equal when transported along $\bar{\gamma}$. By computation for function extensionality, when applied to a, $\bar{\gamma}$ becomes equal to $\mathsf{isotoid}(\gamma_a)$. But by Lemma 9.9.5, transporting $Ff : \hom_B(Fa, Fb)$ along $\mathsf{isotoid}(\gamma_a)$ and $\mathsf{isotoid}(\gamma_b)$ is equal to the composite $\gamma_b \circ Ff \circ (\gamma_a)^{-1}$, which by naturality of γ is equal to Gf.

This completes the definition of a function $(F \cong G) \to (F = G)$. Now consider the composite

$$(F = G) \to (F \cong G) \to (F = G).$$

Since hom-sets are sets, their identity types are mere propositions, so to show that two identities $p, q : F = G$ are equal, it suffices to show that $p =_{F_0 = G_0} q$. But in the definition of $\bar{\gamma}$, if γ were of the form $\mathsf{idtoiso}(p)$, then γ_a would be equal

to $\mathsf{idtoiso}(p_a)$ (this can easily be proved by induction on p). Thus, $\mathsf{isotoid}(\gamma_a)$ would be equal to p_a, and so by function extensionality we would have $\bar{\gamma} = p$, which is what we need.

Finally, consider the composite

$$(F \cong G) \to (F = G) \to (F \cong G).$$

Since identity of natural transformations can be tested componentwise, it suffices to show that for each a we have $\mathsf{idtoiso}(\bar{\gamma})_a = \gamma_a$. But as observed above, we have $\mathsf{idtoiso}(\bar{\gamma})_a = \mathsf{idtoiso}((\bar{\gamma})_a)$, while $(\bar{\gamma})_a = \mathsf{isotoid}(\gamma_a)$ by computation for function extensionality. Since $\mathsf{isotoid}$ and $\mathsf{idtoiso}$ are inverses, we have $\mathsf{idtoiso}(\bar{\gamma})_a = \gamma_a$ as desired. □

Definition 9.9.12. A functor $F : A \to B$ is an **equivalence of (pre)categories** if it is a left adjoint for which η and ϵ are isomorphisms. We write $A \simeq B$ for the type of equivalences of (pre)categories from A to B.

Definition 9.9.13. We say a functor $F : A \to B$ is **faithful** if for all $a, b : A$, the function

$$F_{a,b} : \hom_A(a, b) \to \hom_B(Fa, Fb)$$

is injective, and **full** if for all $a, b : A$ this function is surjective. If it is both (hence each $F_{a,b}$ is an equivalence) we say F is **fully faithful**.

Definition 9.9.14. We say a functor $F : A \to B$ is **split essentially surjective** if for all $b : B$ there exists an $a : A$ such that $Fa \cong b$.

Definition 9.9.15. A functor $F : A \to B$ is **essentially surjective** if for all $b : B$, there *merely* exists an $a : A$ such that $Fa \cong b$. We say F is a **weak equivalence** if it is fully faithful and essentially surjective.

Lemma 9.9.16. *If $F : A \to B$ is fully faithful and A is a category, then for any $b : B$ the type $\sum_{(a:A)}(Fa \cong b)$ is a mere proposition. Hence a functor between categories is an equivalence if and only if it is a weak equivalence.*

证明. Suppose given (a, f) and (a', f') in $\sum_{(a:A)}(Fa \cong b)$. Then ${f'}^{-1} \circ f$ is an isomorphism $Fa \cong Fa'$. Since F is fully faithful, we have $g : a \cong a'$ with $Fg = {f'}^{-1} \circ f$. And since A is a category, we have $p : a = a'$ with $\mathsf{idtoiso}(p) = g$. Now $Fg = {f'}^{-1} \circ f$ implies $((F_0)(p))_*(f) = f'$, hence (by the characterization of equalities in dependent pair types) $(a, f) = (a', f')$.

Thus, for fully faithful functors whose domain is a category, essential surjectivity is equivalent to split essential surjectivity, and so being a weak equivalence is equivalent to being an equivalence. □

Definition 9.9.17. A functor $F : A \to B$ is an **isomorphism of (pre)categories** if F is fully faithful and $F_0 : A_0 \to B_0$ is an equivalence of types.

Lemma 9.9.18. *For categories A and B, a functor $F : A \to B$ is an equivalence of categories if and only if it is an isomorphism of categories.*

证明. Since both are mere properties, it suffices to show they are logically equivalent. So first suppose F is an equivalence of categories, with (G, η, ϵ) given. We have already seen that F is fully faithful. By Theorem 9.9.11, the natural isomorphisms η and ϵ yield identities $1_A = GF$ and $FG = 1_B$, hence in particular identities $\mathsf{id}_A = G_0 \circ F_0$ and $F_0 \circ G_0 = \mathsf{id}_B$. Thus, F_0 is an equivalence of types.

Conversely, suppose F is fully faithful and F_0 is an equivalence of types, with inverse G_0, say. Then for each $b : B$ we have $G_0 b : A$ and an identity $FGb = b$, hence an isomorphism $FGb \cong b$. Thus, by original lemma 9.4.5, F is an equivalence of categories. □

Lemma 9.9.19. *If A, B, C are precategories and $H : A \to B$ is an essentially surjective functor, then $(- \circ H) : C^B \to C^A$ is faithful.*

证明. Let $F, G : B \to C$, and $\gamma, \delta : F \to G$ be such that $\gamma H = \delta H$; we must show $\gamma = \delta$. Thus let $b : B$; we want to show $\gamma_b = \delta_b$. This is a mere proposition, so since H is essentially surjective, we may assume given an $a : A$ and an isomorphism $f : Ha \cong b$. But now we have

$$\gamma_b = G(f) \circ \gamma_{Ha} \circ F(f^{-1}) = G(f) \circ \delta_{Ha} \circ F(f^{-1}) = \delta_b. \qquad \square$$

Lemma 9.9.20. *If A, B, C are precategories and $H : A \to B$ is essentially surjective and full, then $(- \circ H) : C^B \to C^A$ is fully faithful.*

证明. It remains to show fullness. Thus, let $F, G : B \to C$ and $\gamma : FH \to GH$. We claim that for any $b : B$, the type

$$\sum_{(g : \hom_C(Fb, Gb))} \prod_{(a:A)} \prod_{(f : Ha \cong b)} (\gamma_a = Gf^{-1} \circ g \circ Ff) \tag{9.9.21}$$

is contractible. Since contractibility is a mere property, and H is essentially surjective, we may assume given $a_0 : A$ and $h : Ha_0 \cong b$.

Now take $g :\equiv Gh \circ \gamma_{a_0} \circ Fh^{-1}$. Then given any other $a : A$ and $f : Ha \cong b$, we must show $\gamma_a = Gf^{-1} \circ g \circ Ff$. Since H is full, there merely exists a morphism $k : \hom_A(a, a_0)$ such that $Hk = h^{-1} \circ f$. And since our goal is a mere

proposition, we may assume given some such k. Then we have

$$\begin{aligned}\gamma_a &= GHk^{-1} \circ \gamma_{a_0} \circ FHk \\ &= Gf^{-1} \circ Gh \circ \gamma_{a_0} \circ Fh^{-1} \circ Ff \\ &= Gf^{-1} \circ g \circ Ff.\end{aligned}$$

Thus, (9.9.21) is inhabited. It remains to show it is a mere proposition. Let $g, g' : \hom_C(Fb, Gb)$ be such that for all $a : A$ and $f : Ha \cong b$, we have both $(\gamma_a = Gf^{-1} \circ g \circ Ff)$ and $(\gamma_a = Gf^{-1} \circ g' \circ Ff)$. The dependent product types are mere propositions, so all we have to prove is $g = g'$. But this is a mere proposition, so we may assume $a_0 : A$ and $h : Ha_0 \cong b$, in which case we have

$$g = Gh \circ \gamma_{a_0} \circ Fh^{-1} = g'.$$

This proves that (9.9.21) is contractible for all $b : B$. Now we define $\delta : F \to G$ by taking δ_b to be the unique g in (9.9.21) for that b. To see that this is natural, suppose given $f : \hom_B(b, b')$; we must show $Gf \circ \delta_b = \delta_{b'} \circ Ff$. As before, we may assume $a : A$ and $h : Ha \cong b$, and likewise $a' : A$ and $h' : Ha' \cong b'$. Since H is full as well as essentially surjective, we may also assume $k : \hom_A(a, a')$ with $Hk = h'^{-1} \circ f \circ h$.

Since γ is natural, $GHk \circ \gamma_a = \gamma_{a'} \circ FHk$. Using the definition of δ, we have

$$\begin{aligned}Gf \circ \delta_b &= Gf \circ Gh \circ \gamma_a \circ Fh^{-1} \\ &= Gh' \circ GHk \circ \gamma_a \circ Fh^{-1} \\ &= Gh' \circ \gamma_{a'} \circ FHk \circ Fh^{-1} \\ &= Gh' \circ \gamma_{a'} \circ Fh'^{-1} \circ Ff \\ &= \delta_{b'} \circ Ff.\end{aligned}$$

Thus, δ is natural. Finally, for any $a : A$, applying the definition of δ_{Ha} to a and 1_a, we obtain $\gamma_a = \delta_{Ha}$. Hence, $\delta \circ H = \gamma$. □

Theorem 9.9.22. *If A, B are precategories, C is a category, and $H : A \to B$ is a weak equivalence, then $(- \circ H) : C^B \to C^A$ is an isomorphism.*

证明. By Theorem 9.9.11, C^B and C^A are categories. Thus, by Lemma 9.9.18 it will suffice to show that $(- \circ H)$ is an equivalence. But since we know from the preceding two lemmas that it is fully faithful, by Lemma 9.9.16 it will suffice to show that it is essentially surjective. Thus, suppose $F : A \to C$; we want there to merely exist a $G : B \to C$ such that $GH \cong F$.

For each $b : B$, let X_b be the type whose elements consist of:

(i) An element $c : C$; and

(ii) For each $a : A$ and $h : Ha \cong b$, an isomorphism $k_{a,h} : Fa \cong c$; such that

(iii) For each (a, h) and (a', h') as in (ii) and each $f : \hom_A(a, a')$ such that $h' \circ Hf = h$, we have $k_{a',h'} \circ Ff = k_{a,h}$.

We claim that for any $b : B$, the type X_b is contractible. As this is a mere proposition, we may assume given $a_0 : A$ and $h_0 : Ha_0 \cong b$. Let $c^0 :\equiv Fa_0$. Next, given $a : A$ and $h : Ha \cong b$, since H is fully faithful there is a unique isomorphism $g_{a,h} : a \to a_0$ with $Hg_{a,h} = {h_0}^{-1} \circ h$; define $k^0_{a,h} :\equiv Fg_{a,h}$. Finally, if $h' \circ Hf = h$, then ${h_0}^{-1} \circ h' \circ Hf = {h_0}^{-1} \circ h$, hence $g_{a',h'} \circ f = g_{a,h}$ and thus $k^0_{a',h'} \circ Ff = k^0_{a,h}$. Therefore, X_b is inhabited.

Now suppose given another $(c^1, k^1) : X_b$. Then $k^1_{a_0,h_0} : c^0 \equiv Fa_0 \cong c^1$. *Since C is a category, we have* $p : c^0 = c^1$ *with* $\mathsf{idtoiso}(p) = k^1_{a_0,h_0}$. And for any $a : A$ and $h : Ha \cong b$, by (iii) for (c^1, k^1) with $f :\equiv g_{a,h}$, we have

$$k^1_{a,h} = k^1_{a_0,h_0} \circ k^0_{a,h} = p_*\big(k^0_{a,h}\big)$$

This gives the requisite data for an equality $(c^0, k^0) = (c^1, k^1)$, completing the proof that X_b is contractible.

Now since X_b is contractible for each b, the type $\prod_{(b:B)} X_b$ is also contractible. In particular, it is inhabited, so we have a function assigning to each $b : B$ a c and a k. Define $G_0(b)$ to be this c; this gives a function $G_0 : B_0 \to C_0$.

Next we need to define the action of G on morphisms. For each $b, b' : B$ and $f : \hom_B(b, b')$, let Y_f be the type whose elements consist of:

(iv) A morphism $g : \hom_C(Gb, Gb')$, such that

(v) For each $a : A$ and $h : Ha \cong b$, and each $a' : A$ and $h' : Ha' \cong b'$, and any $\ell : \hom_A(a, a')$, we have

$$(h' \circ H\ell = f \circ h) \to (k_{a',h'} \circ F\ell = g \circ k_{a,h}).$$

We claim that for any b, b' and f, the type Y_f is contractible. As this is a mere proposition, we may assume given $a_0 : A$ and $h_0 : Ha_0 \cong b$, and each $a'_0 : A$ and $h'_0 : Ha'_0 \cong b'$. Then since H is fully faithful, there is a unique $\ell_0 : \hom_A(a_0, a'_0)$ such that $h'_0 \circ H\ell_0 = f \circ h_0$. Define $g_0 :\equiv k_{a'_0,h'_0} \circ F\ell_0 \circ (k_{a_0,h_0})^{-1}$.

Now for any a, h, a', h', and ℓ such that $(h' \circ H\ell = f \circ h)$, we have $h^{-1} \circ h_0 : Ha_0 \cong Ha$, hence there is a unique $m : a_0 \cong a$ with $Hm = h^{-1} \circ h_0$ and hence $h \circ Hm = h_0$. Similarly, we have a unique $m' : a'_0 \cong a'$ with $h' \circ Hm' = h'_0$. Now

by (iii), we have $k_{a,h} \circ Fm = k_{a_0,h_0}$ and $k_{a',h'} \circ Fm' = k_{a'_0,h'_0}$. We also have

$$\begin{aligned} Hm' \circ H\ell_0 &= (h')^{-1} \circ h'_0 \circ H\ell_0 \\ &= (h')^{-1} \circ f \circ h_0 \\ &= (h')^{-1} \circ f \circ h \circ h^{-1} \circ h_0 \\ &= H\ell \circ Hm \end{aligned}$$

and hence $m' \circ \ell_0 = \ell \circ m$ since H is fully faithful. Finally, we can compute

$$\begin{aligned} g_0 \circ k_{a,h} &= k_{a'_0,h'_0} \circ F\ell_0 \circ (k_{a_0,h_0})^{-1} \circ k_{a,h} \\ &= k_{a'_0,h'_0} \circ F\ell_0 \circ Fm^{-1} \\ &= k_{a'_0,h'_0} \circ (Fm')^{-1} \circ F\ell \\ &= k_{a',h'} \circ F\ell. \end{aligned}$$

This completes the proof that Y_f is inhabited. To show it is contractible, since hom-sets are sets, it suffices to take another $g_1 : \hom_C(Gb, Gb')$ satisfying (v) and show $g_0 = g_1$. However, we still have our specified $a_0, h_0, a'_0, h'_0, \ell_0$ around, and (v) implies both g_0 and g_1 must be equal to $k_{a'_0,h'_0} \circ F\ell_0 \circ (k_{a_0,h_0})^{-1}$.

This completes the proof that Y_f is contractible for each $b, b' : B$ and $f : \hom_B(b, b')$. Therefore, there is a function assigning to each such f its unique inhabitant; denote this function $G_{b,b'} : \hom_B(b, b') \to \hom_C(Gb, Gb')$. The proof that G is a functor is straightforward; in each case we can choose a, h and apply (v).

Finally, for any $a_0 : A$, defining $c :\equiv Fa_0$ and $k_{a,h} :\equiv Fg$, where $g : \hom_A(a, a_0)$ is the unique isomorphism with $Hg = h$, gives an element of X_{Ha_0}. Thus, it is equal to the specified one; hence $GHa = Fa$. Similarly, for $f : \hom_A(a_0, a'_0)$ we can define an element of Y_{Hf} by transporting along these equalities, which must therefore be equal to the specified one. Hence, we have $GH = F$, and thus $GH \cong F$ as desired. □